

MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

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Abstract—We propose MinImageScorer (MIS), an error-driven framework that scores per-object and image difficulty by combining model confidence and localization penalties on training-set failure signals. Validated on MinneApple and Global Wheat Head Detection (GWHD), MIS reliably ranks objects and images by difficulty and identifies domain- and size-level failure patterns without additional information. Score-guided copy-paste outperforms random selection on MinneApple ($\Delta\text{AP}=+0.013$), but both strategies fall below baseline. On GWHD, domain shift renders copy-paste ineffective regardless of selection strategy, establishing MIS as a reliable diagnostic for structural difficulty in agricultural detection.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

I. INTRODUCTION

Agricultural detection benchmarks present structural challenges absent from general-purpose datasets. MinneApple [1] is dominated by densely packed sub-resolution apple instances where failure is driven primarily by scale. GWHD [2] aggregates images from seven geographic domains, causing systematic performance drops on minority-domain images. Copy-paste augmentation yields gains on MS COCO [3], [4], but applied blindly to structurally constrained datasets it risks compounding failure modes: pasting sub-resolution objects amplifies scale noise; pasting cross-domain objects amplifies distributional mismatch.

We introduce **MinImageScorer** (MIS), a per-object difficulty score derived from model failure signals, and show that MIS-guided copy-paste outperforms random selection on MinneApple while both strategies remain below baseline. On GWHD, copy-paste degrades performance regardless of strategy, demonstrating that the dataset’s structural failure mode—not selection quality—determines whether copy-paste can succeed.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ be ground-truth objects in image I and $\mathcal{P}(g)=\{p : \text{IoU}(p, g) \geq \tau\}$ the predictions matched to g , with $c_p=\text{conf}(p)$ and $u_p=\text{IoU}(p, g)$. The *Object Difficulty Score* is:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset, \\ \alpha + \beta, & \text{otherwise,} \end{cases} \quad (1)$$

where α, β weight confidence and localization penalties. Missed objects receive maximum score $\alpha + \beta$. The *Image Difficulty Score* is:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FP rate}(I). \quad (2)$$

Weights are calibrated by maximizing Pearson correlation between S_{img} and per-image miss and FP rates on the training set, requiring no domain-specific prior knowledge.

B. Score-Guided Copy-Paste Pipeline

A baseline YOLOv8s detector is trained, then full-training-set inference yields per-prediction confidence and IoU values from which MIS scores are computed. The dataset is extended by 30%, with 2–8 objects pasted per new image using scale jitter and random flipping [4]. Under *Score CP*, source objects are sampled proportionally to S_{obj} via exponential weighting; under *Random CP*, uniformly. Both conditions use identical hyperparameters.

III. RESULTS AND DISCUSSION

A. Score Validation

Table I(a) shows strong positive correlations ($r > 0.69$) between S_{img} and per-image miss and FP rates, confirming the score reliably captures image-level difficulty. Table I(b) shows that harder GWHD domains (lower AP) carry higher mean S_{img} , matching the known cross-domain failure mode. Table I(c) shows S_{obj} is higher for small objects than medium objects on MinneApple, correctly mirroring AP (0.300 vs. 0.592). MIS therefore recovers both domain- and size-level difficulty using only training-set model failure signals.

B. Augmentation Results

Table II summarises copy-paste outcomes. On MinneApple, Score CP outperforms Random CP across all sub-metrics ($\Delta\text{AP}=+0.013$), confirming that difficulty-weighted sampling improves selection quality. Both strategies nonetheless fall below baseline: boundary artifacts from unblended pastes reduce localization precision, and sub-resolution objects cannot be made more learnable through repetition alone. On GWHD, Score CP and Random CP are statistically indistinguishable and both degrade below baseline. Score-guided sampling concentrates objects from domain-minority sources, amplifying distributional imbalance rather than correcting it. These results

TABLE I
SCORE VALIDATION. (A) PEARSON r OF S_{img} WITH MISS/FP RATES.
(B) GWHD DOMAIN AP VS. MEAN S_{img} . (C) MINNEAPPLE SCORES BY
OBJECT SIZE.

(a) Correlation with detection failures		
Dataset	$r(\text{score, miss})$	$r(\text{score, FP})$
MinneApple	0.698	0.688
GWHD	0.816	0.818

(b) GWHD domain-level difficulty		
Domain	AP	Mean S_{img}
rres_1	0.572	0.200
ethz_1	0.550	0.227
inrae_1	0.523	0.251
arvalis_3	0.466	0.294
usask_1	0.408	0.312
arvalis_1	0.403	0.331
arvalis_2	0.365	0.424

(c) MinneApple object scores by size		
Metric	Small (≤ 32 px)	Medium (≤ 96 px)
AP	0.300	0.592
S_{obj} (mean)	0.378	0.222

TABLE II
COPY-PASTE AUGMENTATION RESULTS.

Dataset	Condition	AP	AP₅₀	AP₇₅	AP_S	AP_M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	0.428	0.759	0.440	0.286	0.581
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	0.484	0.905	0.455	0.098	0.477

confirm that covariate shift between augmented and test images—not selection strategy—governs GWHD performance.

IV. CONCLUSION

MIS reliably quantifies per-object detection difficulty from model failure signals, recovering size- and domain-level structural patterns without privileged information. Score-guided copy-paste outperforms random selection on MinneApple but degrades below baseline on both datasets: sub-resolution scale limits and cross-domain distributional shift are structural constraints that augmentation alone cannot overcome. Copy-paste is beneficial when objects are underrepresented yet visually compatible with the test distribution; it is counterproductive under resolution limits or domain shift. Future work should pursue better boundary blending to reduce distributional displacement and dynamic score recomputation to adapt augmentation as the model improves.

REFERENCES

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